

VAJRA

Visual Audio Joint-modal Rendering Architecture

VAJRA 2.0

Document 4 of 5

Technical Writeup

How the platform works, at the level of mechanism

WHO THIS IS FOR

A researcher or engineer inheriting the work

WHAT IT COVERS

- Architecture: isolation, orchestration, one-model-at-a-time
- The editing pipeline, from transcript to re-synchronised video
- The realism chain and the detectability scorecard
- Where the research contributions sit

THE FULL SET

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|--|---------------------------|
| 1. System Requirements and Sizing | specifying the machine |
| 2. Deployment and Operations Guide | installing and running it |
| 3. Operator's User Manual | using it |
| ■ 4. Technical Writeup | understanding it |
| 5. Verification and Test Documentation | maintaining it |

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Offline synthetic-media platform · for authorised research and capability development

1. What the Platform Is

Seven media-synthesis modules behind one interface, running offline on a single GPU. The modules share an orchestration layer but not a runtime: each model family lives in its own isolated environment and is driven as a subprocess.

The system's distinguishing property is not any one model — the backbones are open models, adapted rather than re-derived. It is that the platform **measures its own output** and reports how detectable an edit is, rather than leaving that to inspection.

2. Architecture

Isolation by model family

The model families require mutually incompatible versions of the tensor, transformer and diffusion libraries. Not inconvenient — actually incompatible. One environment cannot satisfy them.

Each family therefore gets its own virtual environment, and the main process talks to it by launching a subprocess worker with a clean, explicitly constructed environment. The cost is disk and a process boundary to cross. The benefit is that adding or upgrading a model cannot break another, which is what allowed the platform to reach seven modules without regression.

Environment	Carries	Used by
principal runtime	the interface, image generation, face swap, media tools	03, 04, 07
venv_voice	the zero-shot cloning stack	01, 02
venv_latentsync	the diffusion lip-sync engine	01
venv_vitor	the local-infill speech engine	01, 02
venv_demucs	source separation	01, 07
venv_qwen	instruction-based image editing	06
venv_ltx2 / venv_wan	the two video-generation engines	05

One model resident at a time

A single-occupant cache holds one model in GPU memory; loading another evicts the first. A process-wide lock serialises every GPU job. Two large models are therefore never resident together, and a second request queues rather than competing.

This is a deliberate trade of throughput for predictability. Two models in VRAM at once fails late and unpredictably; a queue fails visibly and never corrupts a run. It is also why the platform's VRAM requirement is the largest single module rather than the sum of all seven.

3. The Editing Pipeline

Module 01 is the heart of the system. Its problem statement: change what a person appears to say, while regenerating as little as possible.

- 1. Transcribe.** Word-level timings are recovered, not just segment text — everything downstream depends on knowing when each word was said.
- 2. Diff.** The edited transcript is compared with the original at word level to find the minimal changed spans.
- 3. Route.** A four-tier router picks the least invasive method the language and installation allow, and records what it chose and why.

- 4. Synthesise.** Only the changed span is regenerated, in the speaker's voice, using the surrounding audio as context where the tier allows.
- 5. Fit.** The replacement is fitted to the time it must occupy — absorbing neighbouring silence first, then time-stretching within a bounded limit, then escalating to a whole line rather than truncating.
- 6. Match and splice.** Five properties of the replacement are matched to its surroundings before it is spliced in at a zero crossing.
- 7. Re-synchronise.** Only the video windows spanning an edit are re-rendered; the rest of the footage is untouched.
- 8. Score.** The result is measured against the recording's own statistics and reported.

The tiers

Tier	Method	What survives
A	In-place infill using the surrounding recording as context	Everything except the changed words
B	In-place infill, non-commercially licensed engine	As above; off by default
C	A pause-bounded phrase is regenerated	Everything outside that phrase
D	The whole transcript line is regenerated	Everything outside that line

The counter-intuitive point, and the reason routing exists at all: **the strongest engine is not the least detectable**. Regenerating more audio than necessary destroys genuine material, and genuine material never needs disguising.

4. The Realism Chain

A synthetic insert differs from its surroundings in measurable ways. Each is corrected in turn, and every stage is **bounded** — it has a cap, and it refuses to act when its own measurement is unreliable.

Property	Why it betrays an edit	Correction
Loudness	A level step is audible at about 1 dB	Matched to the surrounding audio against a broadcast loudness standard
Spectral balance	A different microphone or room has a different timbre	Long-term average spectrum matched toward the neighbourhood
Noise floor	A silent insert is as obvious as a loud one	Room tone injected to the measured level of the surroundings
Seam	A discontinuity at the join is heard as a click	Spliced at a zero crossing with a short crossfade
Duration	The track must stay frame-aligned with the picture	Silence absorbed, then bounded time-stretch, then escalation

The bounding matters more than the correction. A filter that always acts will eventually damage material that was already correct — a lesson learned from a real fault, where the chain applied gain to output that had been produced from that same recording and needed no disguising.

5. The Detectability Scorecard

The measurement problem: *is this edit noticeable?* Judging by ear is not repeatable between two reviewers, and a fixed threshold in decibels is no better, because it says nothing about a particular recording — ten decibels of discontinuity may be unremarkable in one and glaring in another.

The approach is self-referential. The system finds every genuine word boundary in the operator's own recording, measures the discontinuity across each, and builds that distribution. The edit's seam is then reported as a percentile within it. A 40th-percentile result means the splice is smoother than 60% of the real boundaries in that same speaker's audio — a statement that is reproducible, which a bare decibel figure is not.

The recording becomes its own control group. No external reference, no calibration, no threshold to argue about.

What it deliberately does not claim

It measures whether the **join** is detectable, not whether the replacement sounds good. A perfectly blended splice of poor synthesis still passes. Four metrics are named as unmeasured — speaker similarity, naturalness, intelligibility and lip-sync accuracy — because omitting them would imply a completeness the system cannot support.

That limitation was found in trial rather than in theory: an edit that scored a pass was audibly over-processed, and the fault lay inside the span, where the scorecard has no metric.

6. Windowed Re-Synchronisation

Lip-sync is confined to the windows containing an edit. Frames with no reason to change are not regenerated, so the great majority of the footage remains the original recording. On a ten-minute clip with a few word changes, under half a percent of frames are re-rendered.

The same principle governs the audio: material outside an edit is not re-encoded, not resampled and not regenerated. It is bit-exact.

7. Where the Research Sits

Contribution	Level	Status
Temporal-consistency objective for GAN talking-face synthesis — feature smoothness plus optical-flow-guided warping	a training objective	accepted, IEEE INCECT
Prosody-gated cross-attention and reference-free valence-arousal conditioning on a diffusion transformer	an architectural addition	under review, INDISCON 2026
Contract-driven orchestration with typed adapters and an end-to-end identity metric across a multi-model chain	a systems contribution	under review, DAWAK 2026

They sit at three deliberately different levels, and the third exists because building the first two inside one platform raised questions the literature had not answered: how to type the interface between two models with different assumptions, and how to measure identity across a chain rather than a single model.

8. Known Limits of This Line

- **Tier-A infill is English only.** No published model provides localised infill for Hindi or code-mixed speech. A field-level gap, not an implementation shortfall.
- **Replacement speech must fit its slot.** Beyond about 15% time-stretch the artefact is audible.
- **One speaker in frame per edit.** The lip-sync models cannot disambiguate speakers.
- **The generated mouth region is lower resolution than the face around it.** Sharper rendering and grain matching are next-phase work.
- **No provenance marking.** Deferred by decision, not slippage: the standard already exists and it can be added late, whereas the realism work could not be deferred because everything else depends on it.

9. Design Rationale Lives in the Code

Every non-obvious decision in this repository is recorded beside the code it governs, not in a separate design document. A design note in a folder goes stale the first time the code changes, because nothing forces anyone to open it; a comment on the function is edited by whoever edits that function.

A maintainer inheriting this work should read the comments in `engines/transcript_engine.py`, `core/router.py` and `dsp/` before changing anything in the editing path. They record why, not what.